



No more Handbrakes

Handbrakes are dying since more and more car manufacturers are ditching them due to the improved car brakes. <https://digit.in/oct18-75-1>



Indian GPS module

India's all set to get its own navigation system up and running after the launch of UtraQ navigation module. <https://digit.in/oct18-75-2>

Launch Vehicle Mark III (GSLV-MKIII). The rocket has a shorter profile than other launch vehicles in ISRO's fleet and measures 43.43 meters in height, but is more stocky. Because of this, ISRO scientists had nicknamed the rocket "Fat Boy". After the first developmental flight on June 5, 2017, the rocket earned the nicknames "Obedient Boy" and "Baahubali". It is a three stage rocket with two strap-on motors, and has the capability of deploying a payload weighing 8,000 KG to the low Earth orbit, which is the target orbit for the first manned spaceflight. The core stage is called the L110, has two liquid propulsion engines. The strap on motors form the second stage, and consist of two solid rocket S200 engines. The third stage is a cryogenic engine. All the three stages are developed indigenously, by the Liquid Propulsion Systems Centre and the Vikram Sarabhai Space Centre. There are two modules for the mission, the crew module and a service module, which will be assembled together to form the orbital module. The service module itself has two liquid propulsion engines, and will provide temperature regulation, pressure control and nutrition services to the crew module. The orbital module will be placed on top of the rocket. The mission is expected to be launched from a new launch pad that is yet to be built at ISRO's launch complex in Sriharikota. This new launch pad is expected to have the capabilities for boarding the crew members into the spacecraft, and the other required equipment for future manned space flights. The flight itself is expected to last around 16 minutes, after which Gaganyaan will be deployed into low Earth orbit at an altitude of 400 kms above the surface of the Earth. The Gaganyaan system is designed to accommodate around three crew members, who will spend between five to seven days in orbit. After the orbital period, the whole orbital module will be reoriented and re-enter the Earth. At this point, the propulsion

system on board the service module will fire the engines for slowing down the descent of the spacecraft. The service module will then separate from the crew module, at an altitude of 120 kms. The parachutes will be deployed for further slowing the crew module down, after which the module will splash into the Arabian Sea. The crew will be recovered off the coast of Gujarat. In case of any issues, there are backup plans to recover the crew from the Bay of Bengal or even on land. The safety of the crew is the top priority for ISRO. There are

take support from other organisations. After they are selected, the crew will undergo a training regimen that is expected to last between two to three years. Given the timelines, the personnel are expected to be selected as early as 2019 itself. As such, the first manned spaceflight is primarily a technology demonstration mission. The scientists and engineers involved will get a first hand experience in planning and executing a manned spaceflight. There is a scientific aspect to the mission though, the astronauts are expected to conduct



Credit: ISRO

Dr Sivan next to the space suit at BSX 2018

two spaceflights planned before the manned mission, to make sure everything goes as expected. The descent is expected to take about 36 minutes. The entire mission will be realised entirely indigenously, and is expected to cost less than ₹10,000 crore.

THE MISSION OBJECTIVES

The crew for the mission will be selected jointly by the Indian Air Force and ISRO. ISRO already has the capabilities for handling the technical aspects of the mission. However, the introduction of manned personnel is something that ISRO does not have experience with, and bioscience is a new field for the space agency. To tackle this aspect of the mission, ISRO hopes to collaborate with and

experiments in microgravity over the duration of the flight. The specific experiments are not mentioned yet, but this is an opportunity for Indian scientists to design and send up self contained experiments into space. The effort will involve academia, private industry partners and various higher educational institutes across India. Realising the mission will also help build capacity for future manned missions. The mission will encourage the growth of industries that are required for more advanced space programs. A successful demonstration would also lead to opportunities for participating in international space programs. One of the most important objectives of the mission is also very simple - inspiring the youth. 